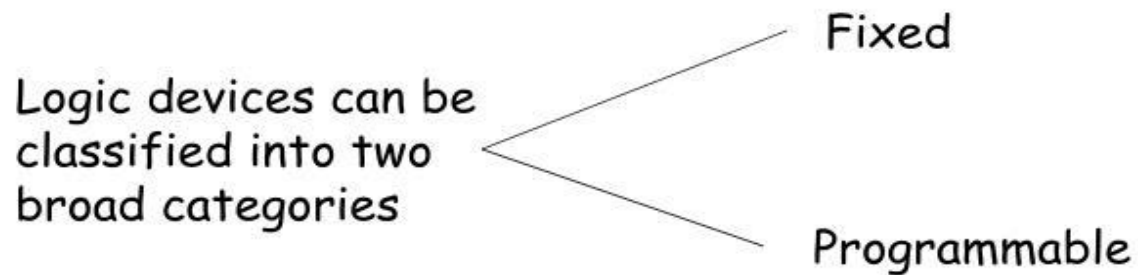


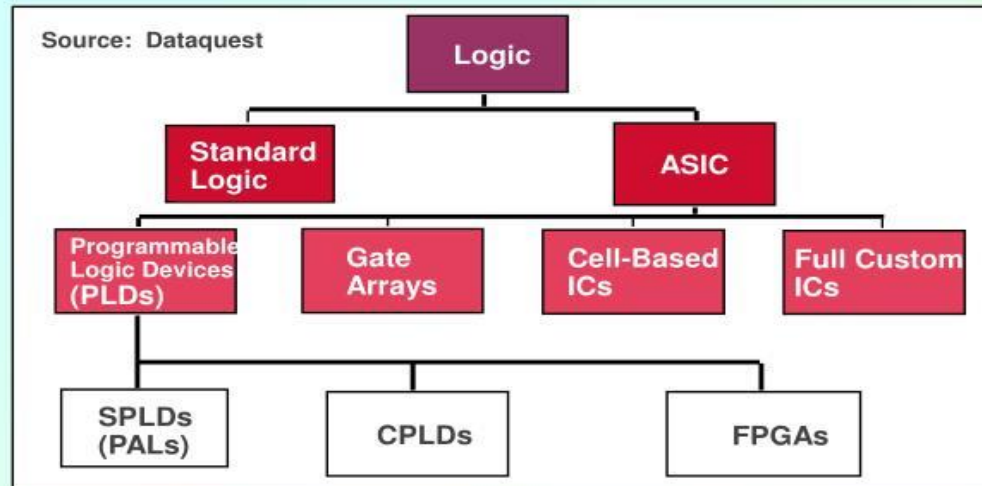
2.5 Programmable Logic Devices

Logic Devices

Programmable Logic Device Introduction



Programmable Logic Device Families



Acronyms

SPLD = Simple Prog. Logic Device

PAL = Prog. Array of Logic

CPLD = Complex PLD

FPGA = Field Prog. Gate Array

Common Resources

Configurable Logic Blocks (CLB)

- Memory Look-Up Table
- AND-OR planes
- Simple gates

Input / Output Blocks (IOB)

- Bidirectional, latches, inverters, pullup/pulldowns

Interconnect or Routing

- Local, internal feedback, and global

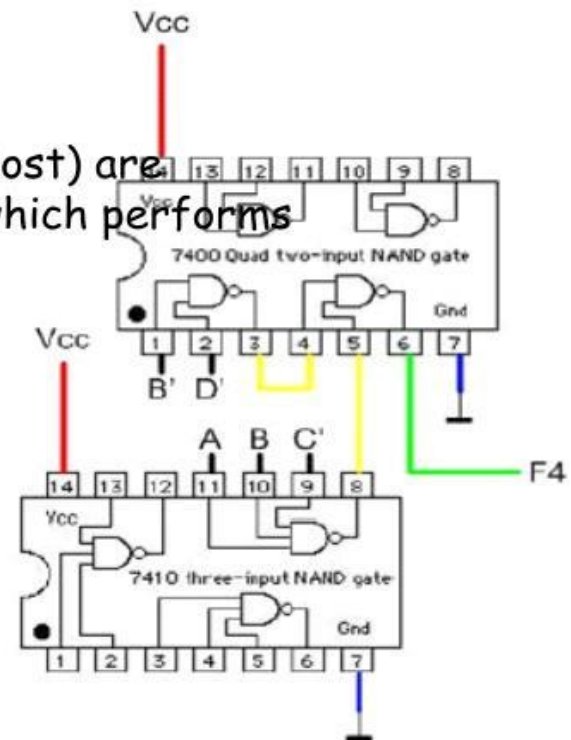
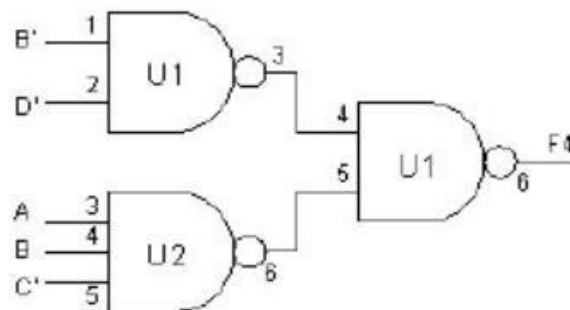
Fixed Logic Devices (e.g. SSI/MSI)

Small-Scale Integration (SSI) uses circuits containing transistors numbering in the tens, while Medium-Scale Integration (MSI) contains hundreds of transistors on each chip.

Disadvantages

- Circuits are permanent
- They perform one function or set of functions
- Once manufactured, they cannot be changed
- Constrained to parts
- Need to stock many different parts
- Most resources (power, board area, manufacturing cost) are consumed by the "package" but not by the "silicon", which performs the actual computation.
- Automation is impossible

Example





Programmable Logic Device

☯ What is a Programmable Logic Device (PLD)?

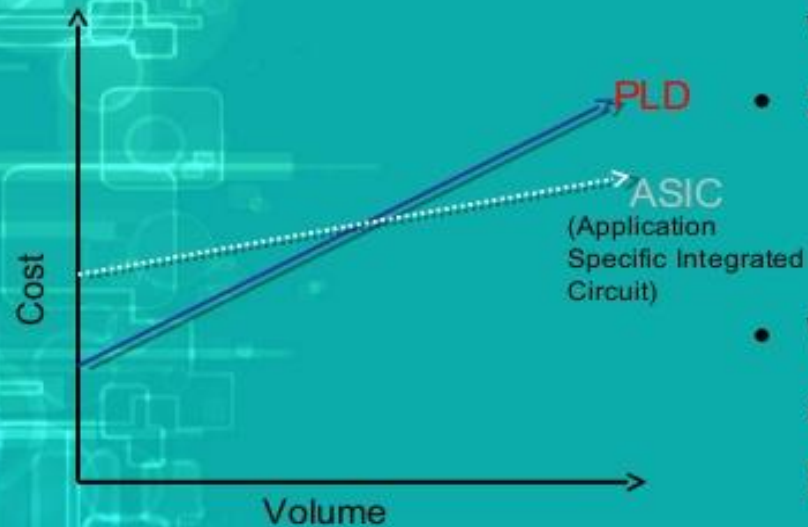
- an IC that contains large numbers of gates, flip-flops and registers that are interconnected on the chip
- can be configured by the user to perform a logic function
- many of the connections are fusible links that can be broken

Purpose of PLDs

- Permits elaborate digital logic designs to be implemented by the user on a single device.
- Is capable of being erased and reprogrammed with a new design.



Advantages of PLDs



- Cost effective in lower volumes
- Short design time
- Well suited for academics and prototyping



Programmable Logic devices

Advantages:

- *Flexibility - reprogramming*
- *Time To Market – TTM*
- *Low cost solution for small quantities*

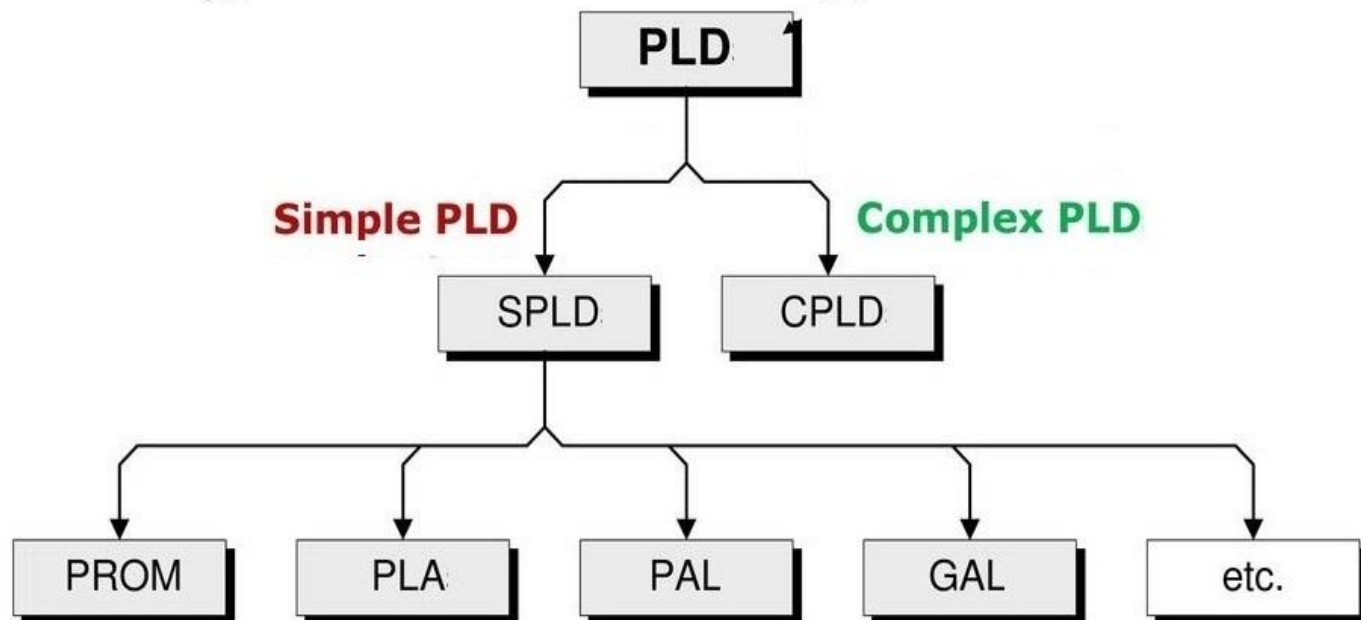
Disadvantages:

- *High cost solution in large quantities*
- *Not so high integration -> large area and poor performance*
- *Not flexible in terms of integrating analog blocks*

PLDs are available in a wide range of sizes up to 10 M gates

These chips consist of a large number of small logic elements, which can be connected together using programmable switches

Programmable Logic Devices



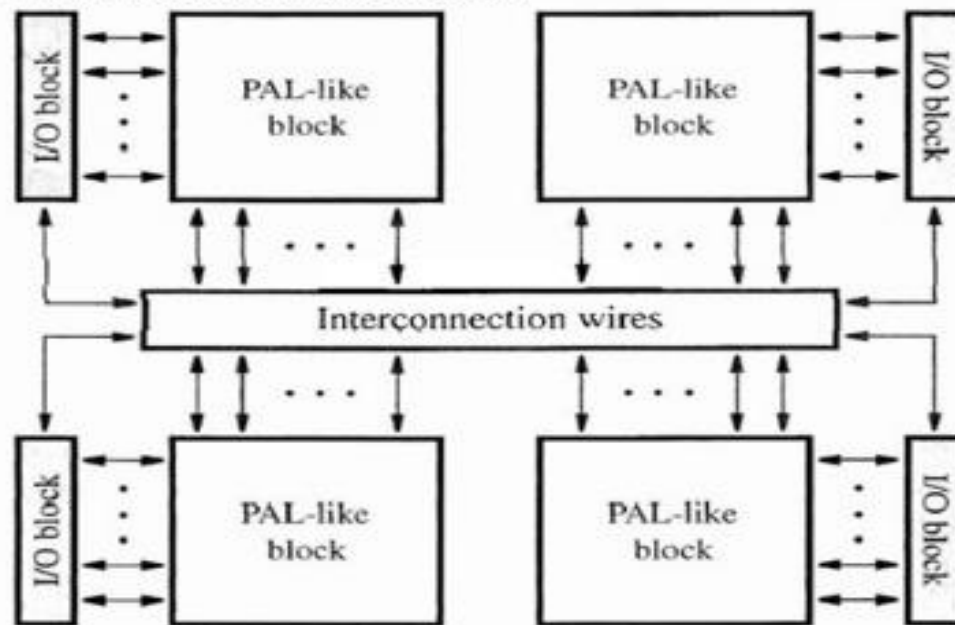
Simple Programmable Logic Devices (SPLDs)

- Simple programmable logic devices (SPLD) are the simplest, smallest and least-expensive forms of programmable logic devices. SPLDs can be used in boards to replace 7400-series TTL components (AND, OR, and NOT gates).
- Most SPLDs use either fuses or non-volatile memory cells (EPROM, EEPROM, FLASH, and others) to define the functionality.



Complex Programmable Logic Devices

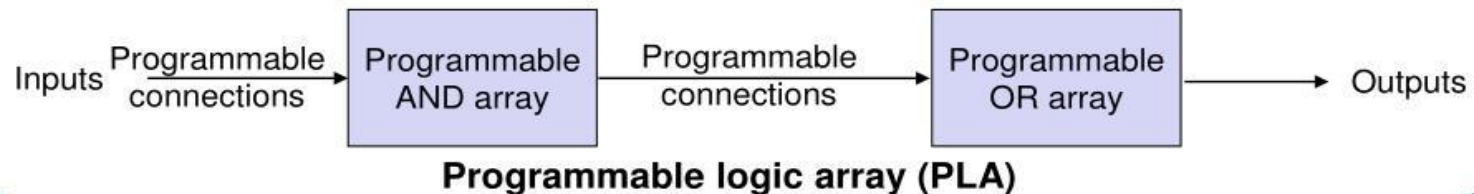
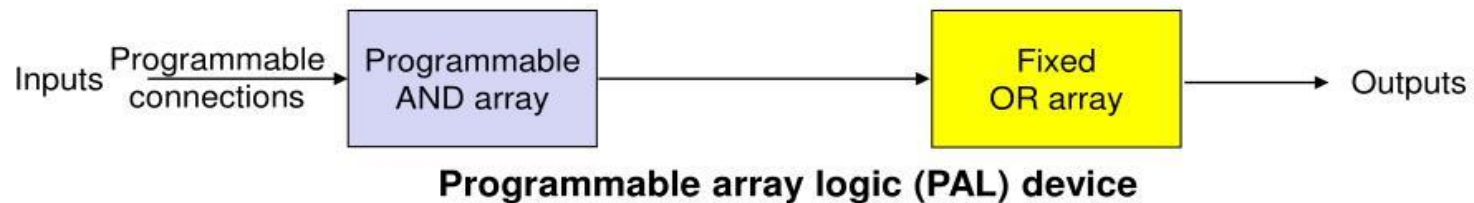
A CPLD comprises multiple circuit blocks on a single chip, with internal wiring resources to connect the circuit blocks. Each circuit block is similar to a PLA or a PAL; we will refer to the circuit blocks as PAL-like blocks. An example of a CPLD is given in Figure below. It includes four PAL-like blocks that are connected to a set of interconnection wires. Each PAL-like block is also connected to a sub circuit labeled I/O block, which is attached to a number of the chip's input and output pins.



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Programmable Logic Devices (PLDs)

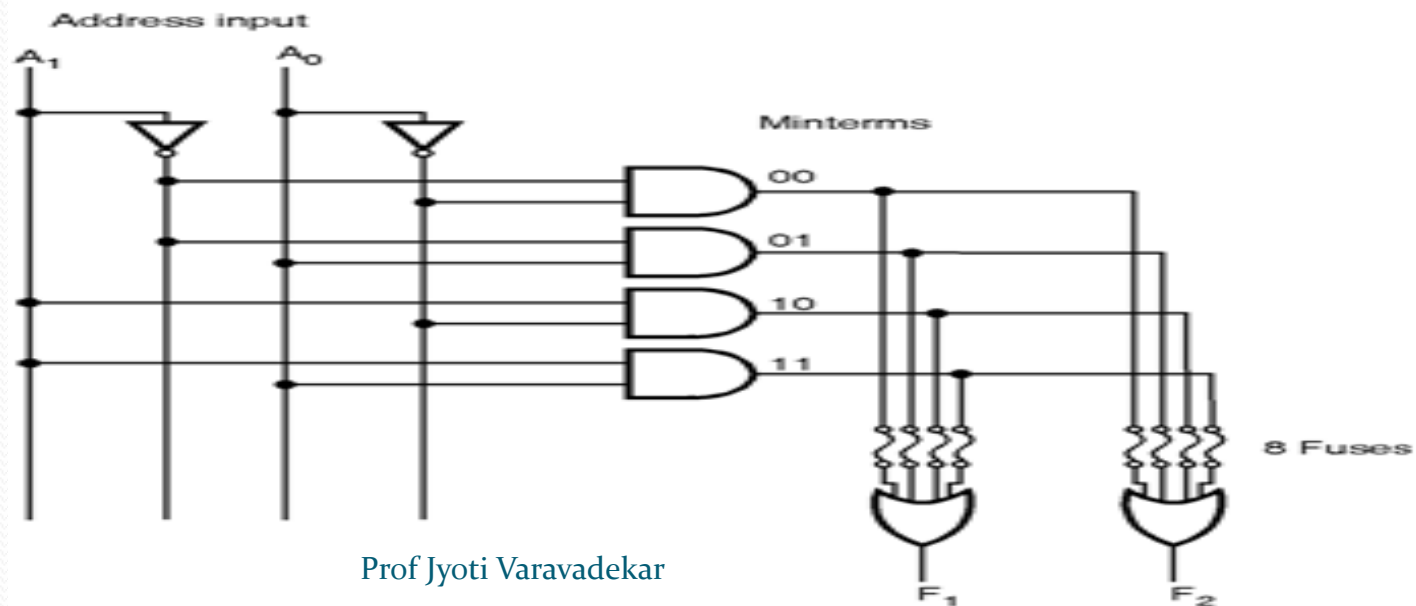
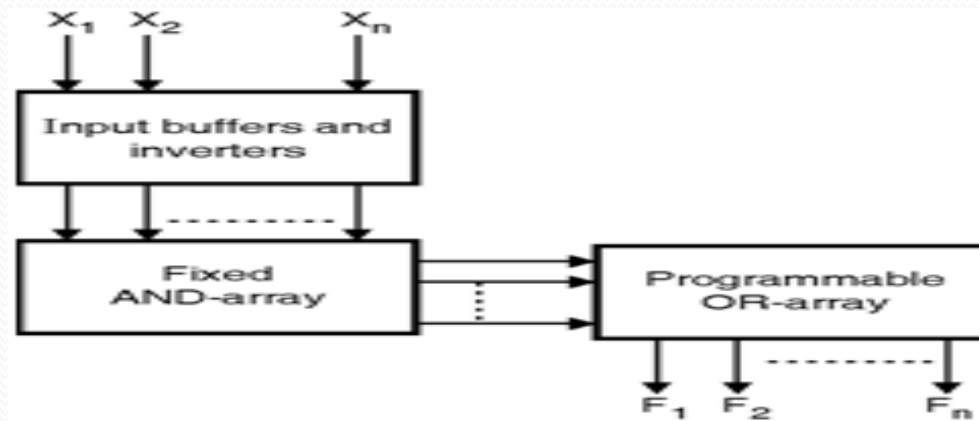
All use AND-OR structure- differ in which is programmable



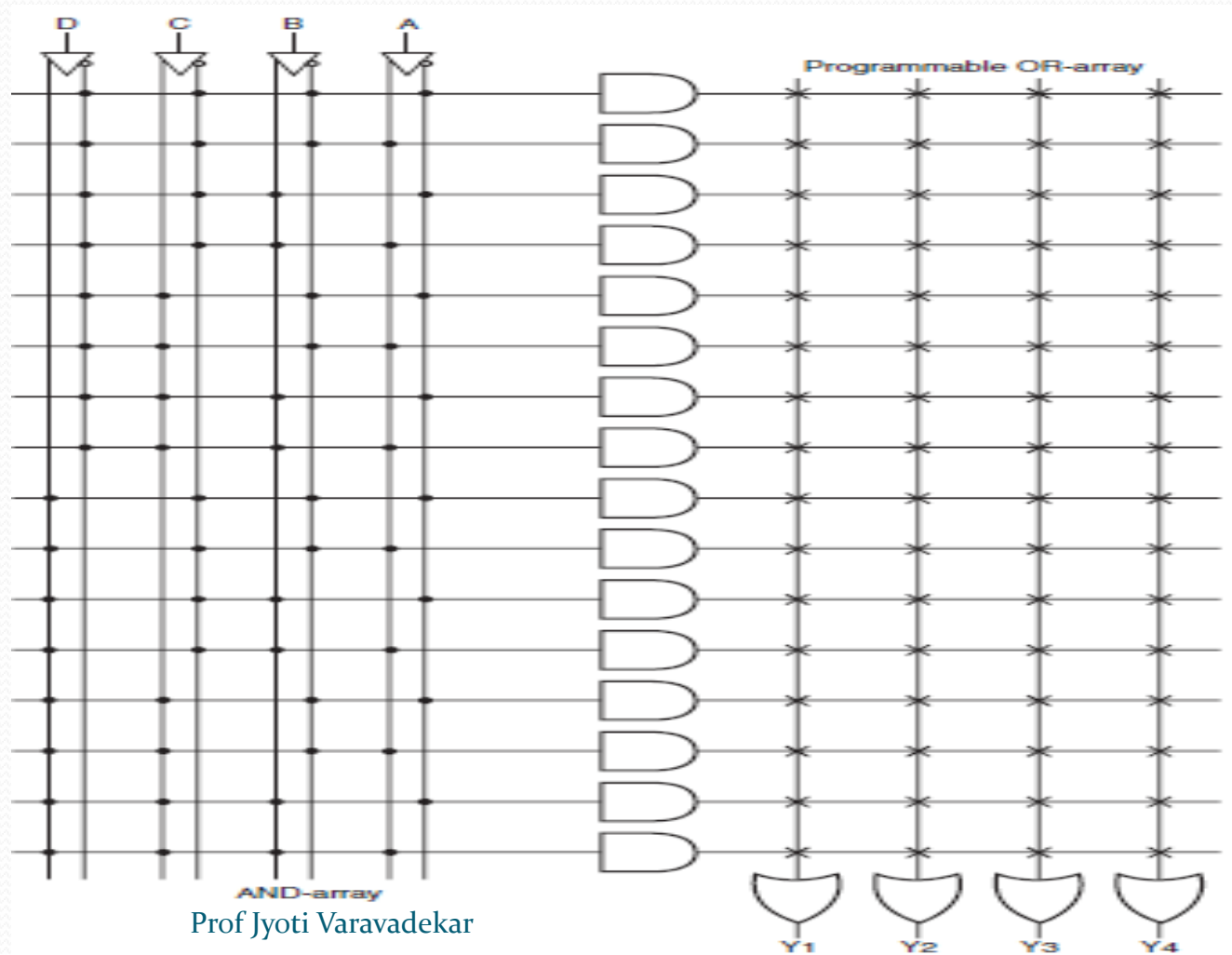
KFUPM

Programmable Read Only Memory (PROM)

- A programmable read only memory is a device that includes both the AND plane and OR-plane within a single IC package. Out of these two arrays AND plane is fixed and OR plane is programmable. Figure below shows the block diagram view of PROM. In the PROM the AND array will act as a decoder which will decode the address lines. The gate level structure of PROM is also shown in figure. In mask PROM it is necessary to specify the bit pattern to be stored according to the requirements of the circuits. Since PROMs are used in logic designs these are also referred as PLDs.



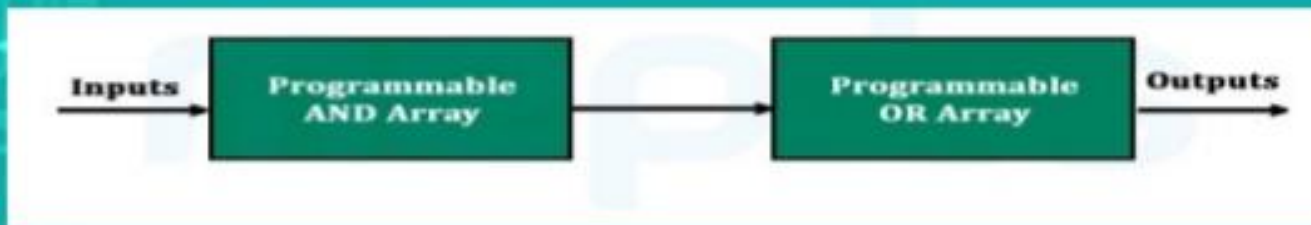
Internal architecture of a PROM having four input lines, a hard-wired array of 16 AND gates and a programmable array of four OR gates.



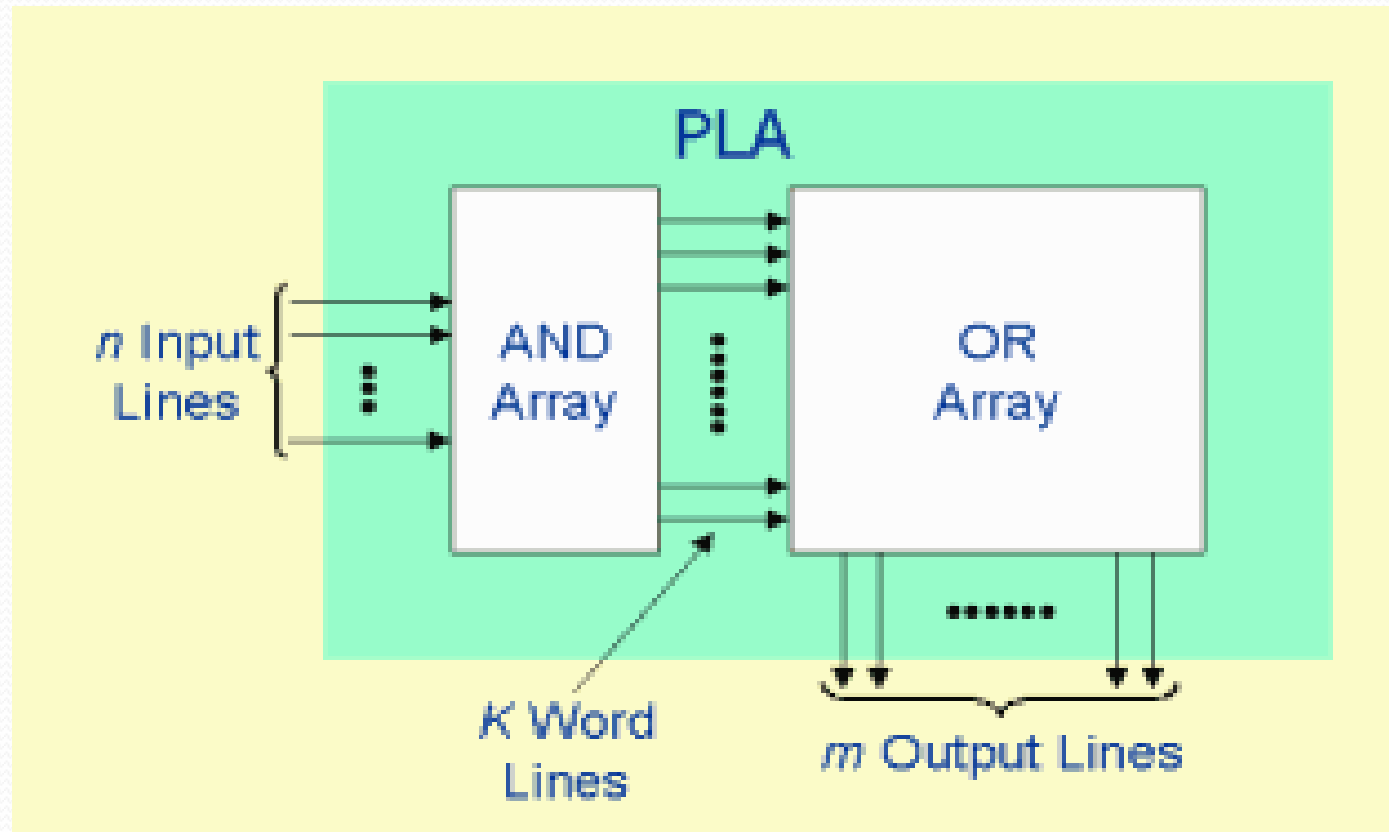
- EPROMs and Electrically Erasable Programmable Read Only Memory (EEPROMs) are programmed through the PROM programmers.
- **Disadvantages of PROM**
- An inefficient use of logic capacity.
- It is costly to use PROMs for applications where lower minterms are needed.
- Higher power consumption

Programmable Logic Arrays (PLAS)

- A **programmable logic array** (PLA) is a kind of **programmable logic** device used to implement combinational **logic** circuits.
- The PLA has a set of **programmable** AND gate planes, which link to a set of **programmable** OR gate planes, which can then be conditionally complemented to produce an output.
- Programmable Logic Array or PLA is used to implement logic functions in digital circuits. The structure has programmable AND-matrix, programmable OR-matrix, input and output buffers. Block diagram of a **PLA** device is as shown below;



PLA Architecture



PLA Logic

Design Example

Multiple functions of A, B, C

$$F1 = A B C$$

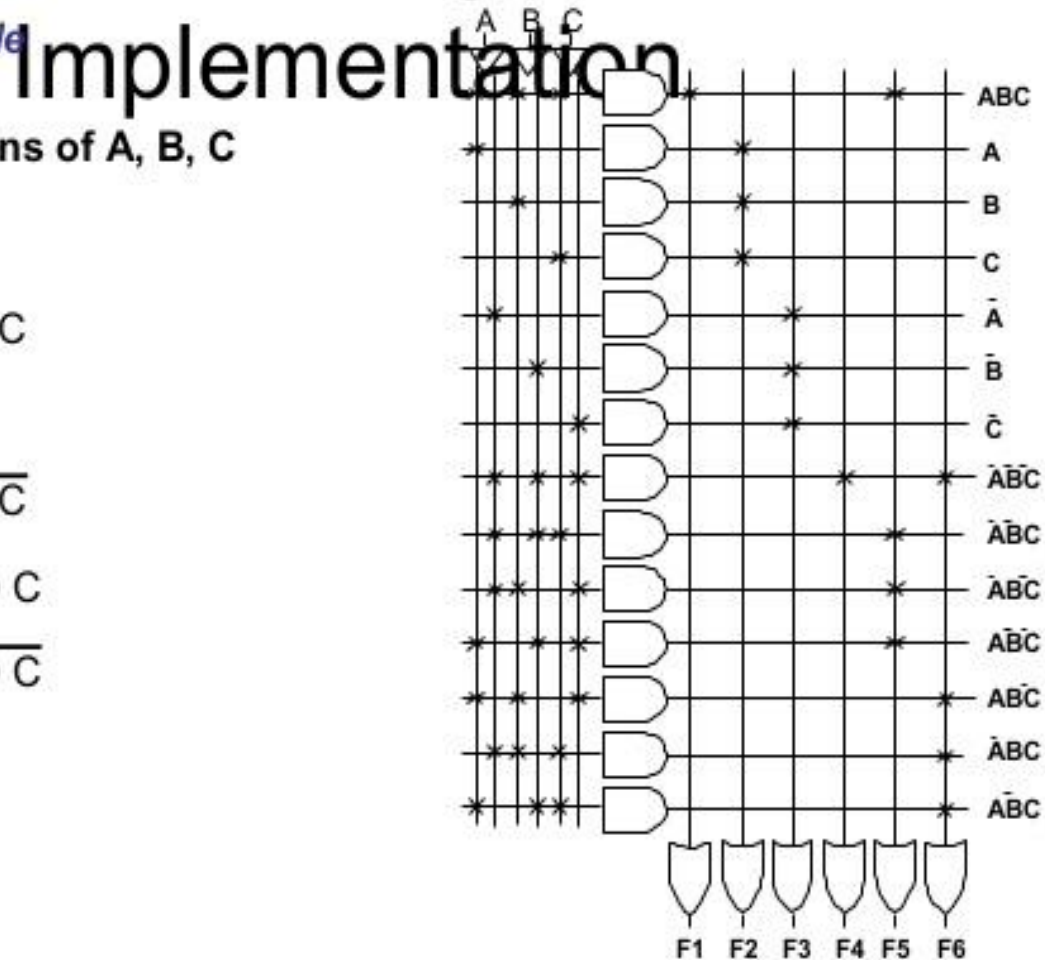
$$F2 = A + B + C$$

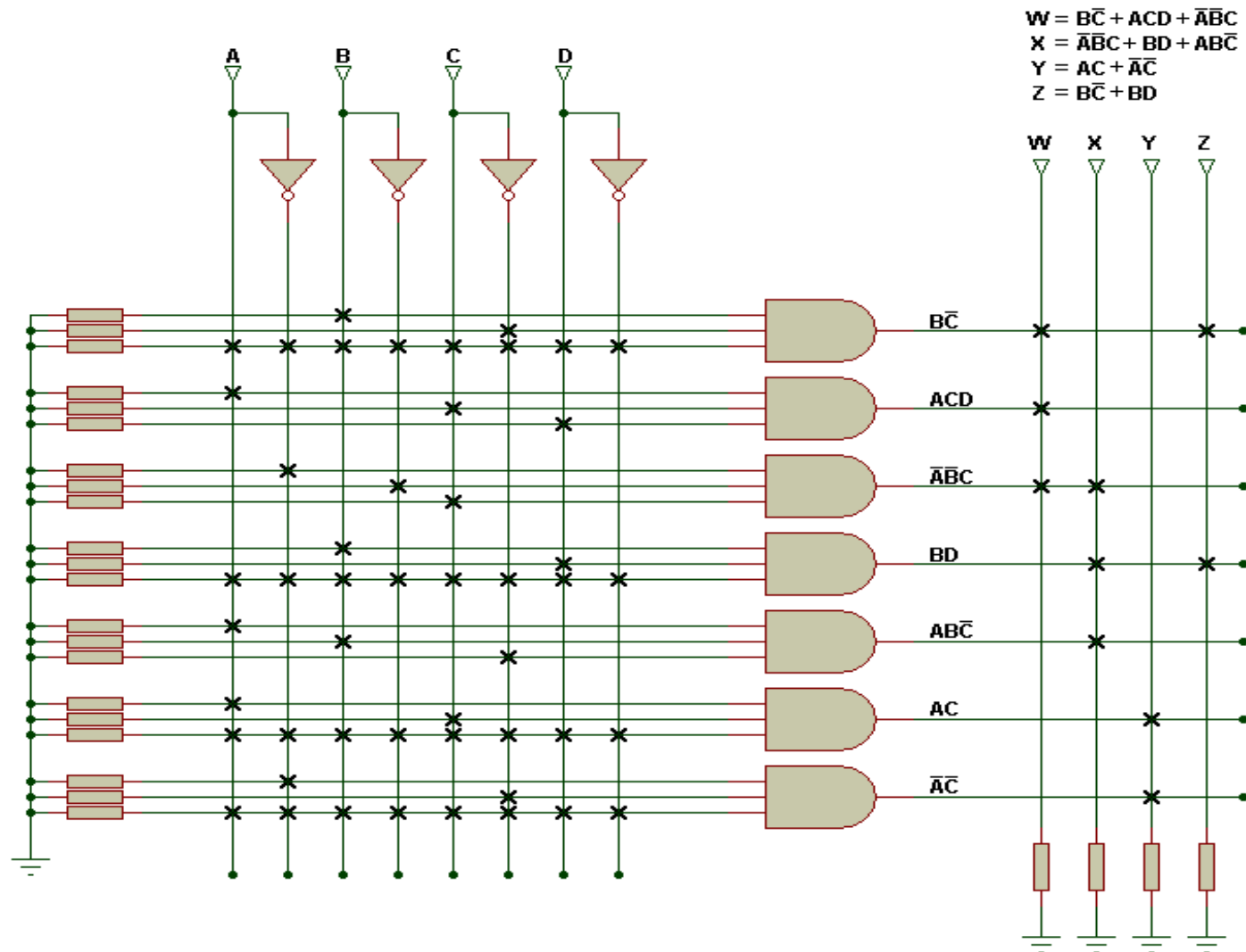
$$F3 = \overline{A B C}$$

$$F4 = \overline{A + B + C}$$

$$F5 = A \oplus B \oplus C$$

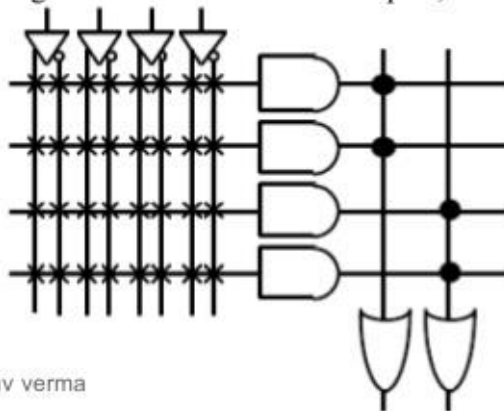
$$F6 = \overline{A \oplus B \oplus C}$$





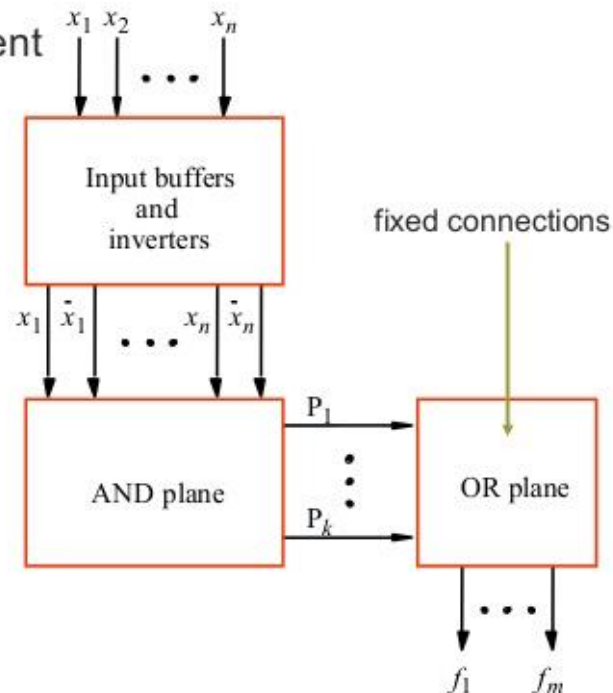
Programmable Array Logic

- PAL is the most commonly used type of PLD. It is a programmable array of logic gates. The array of logic gates is on the single chip and it is in the AND-OR configuration.
- The special feature of PAL is that it has a programmable AND array and a fixed OR array.
- Also note that each OR gate in the OR array gets inputs from some of the AND gates. That means Outputs of all the AND gates are not applied to any of the OR gates. Figure below shows the configuration for a PAL with 4 inputs, 4 AND gate (programmable) and 2 fixed OR gates.



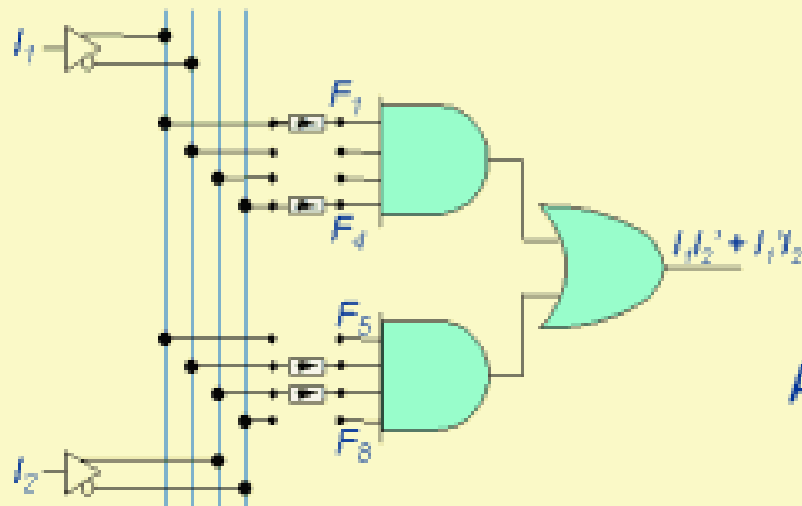
■ Programmable Array Logic (PAL)

- Also used to implement circuits in SOP form
- The connections in the AND plane are programmable
- The connections in the OR plane are NOT programmable



Programmable Array Logic (PAL)

A PAL is a special case of a PLA in which the AND array is programmable but the OR array is fixed.

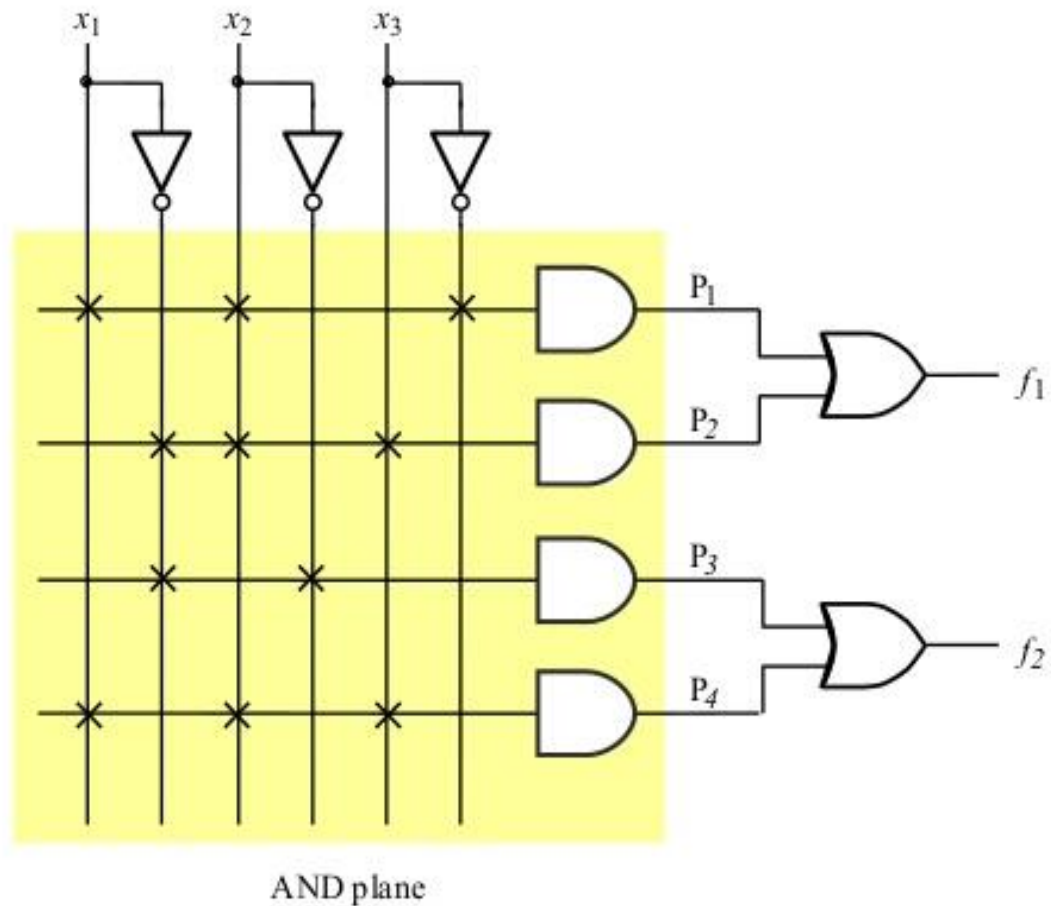


A programmed
PAL

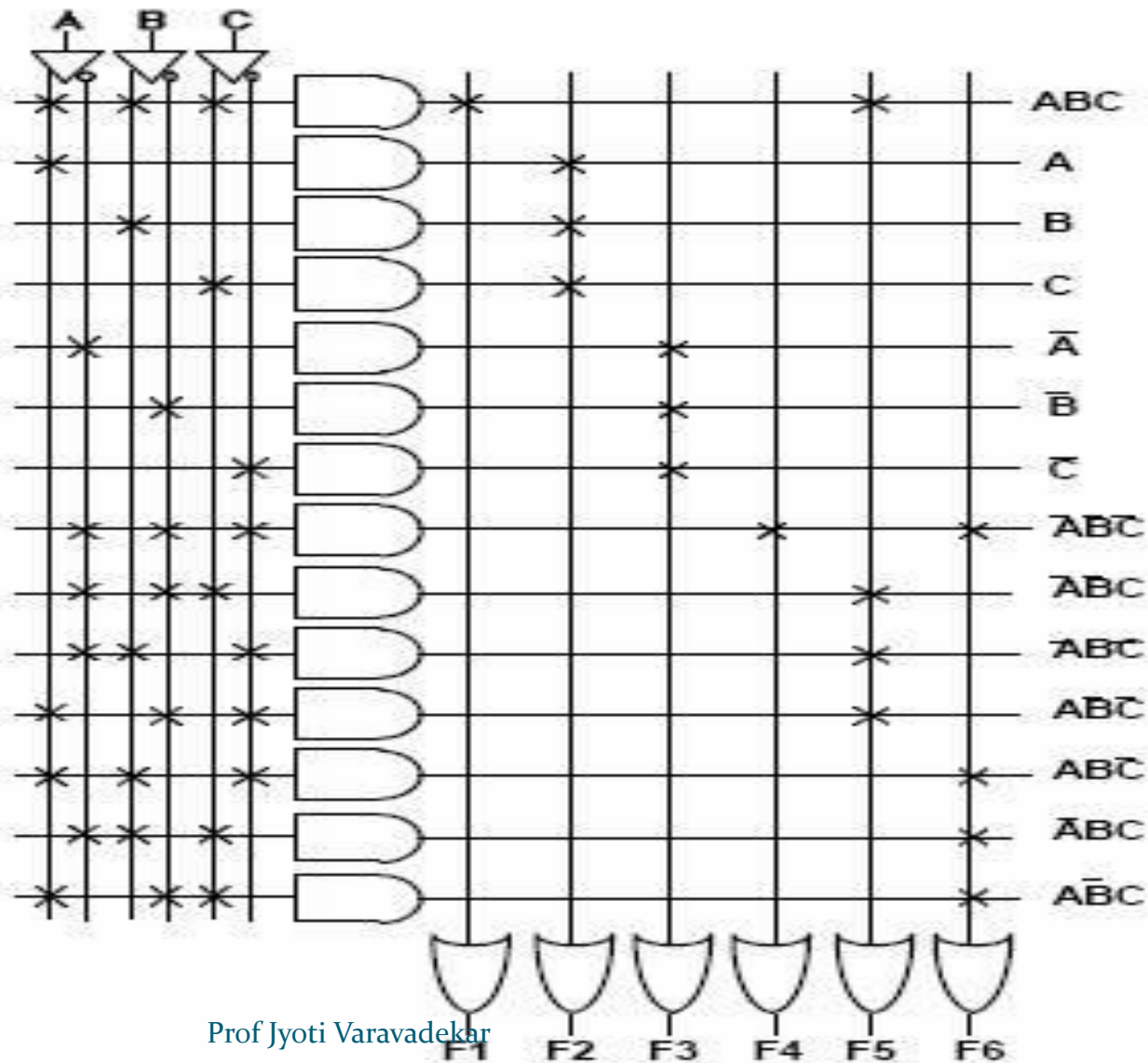
Example Schematic of a PAL

$$f_1 = x_1x_2x_3' + x_1'x_2x_3$$

$$f_2 = x_1'x_2' + x_1x_2x_3$$

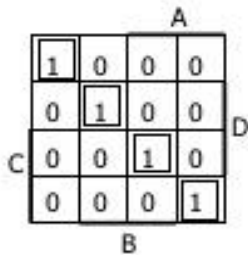


PAL Example

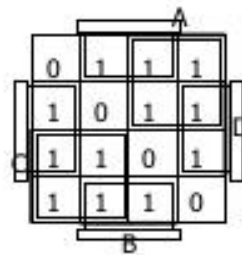


PALs and PLAs: another design example

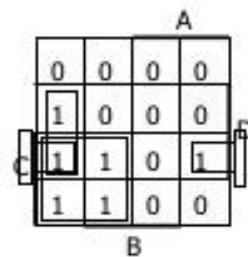
□ Magnitude comparator



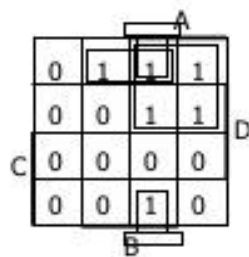
K-map for EQ



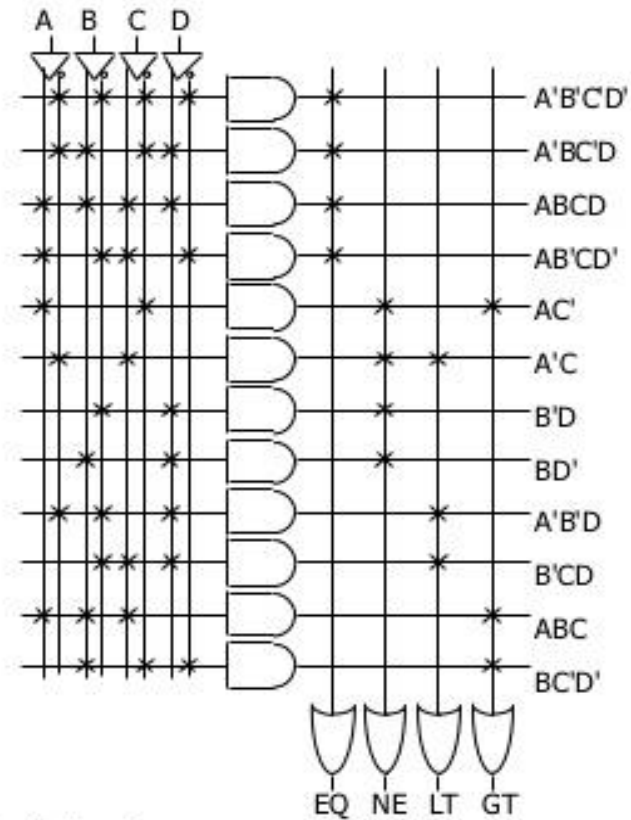
K-map for NE



K-map for LT

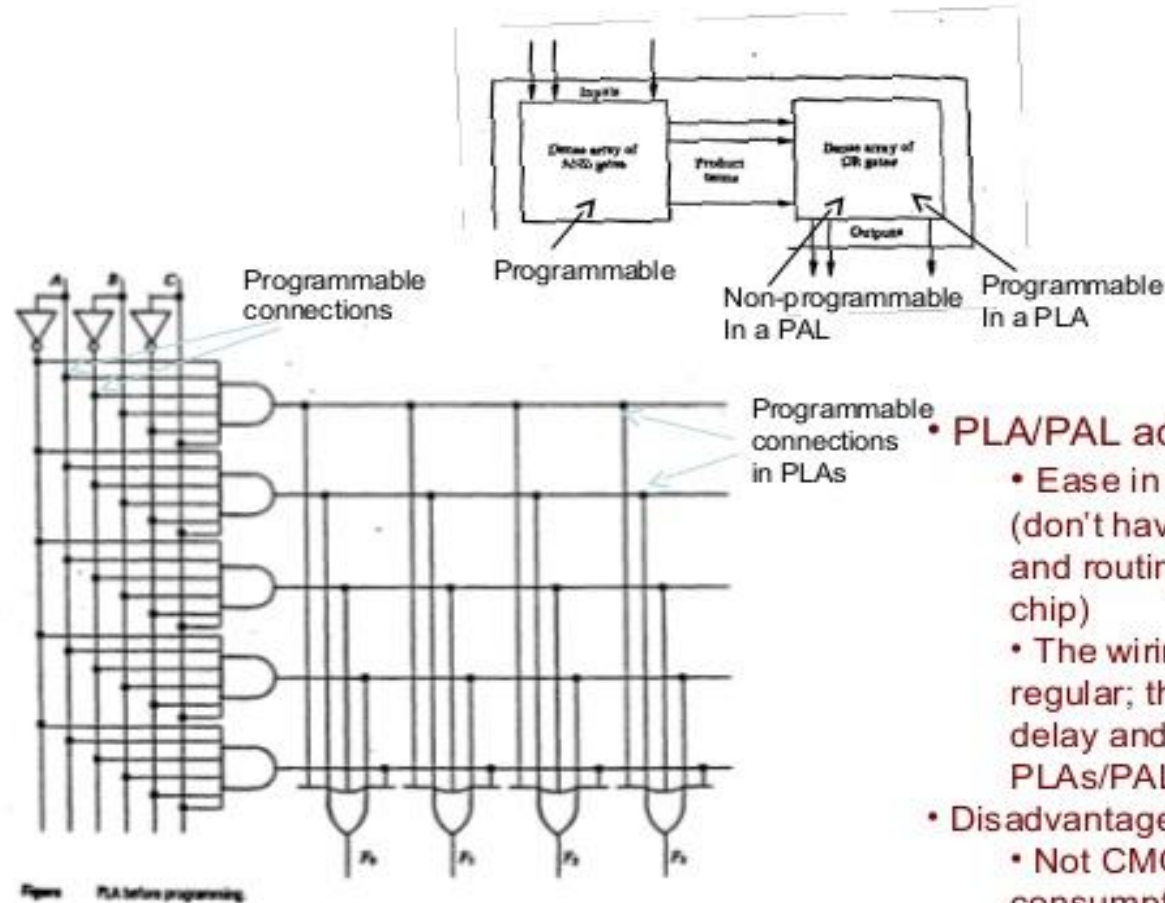


K-map for GT



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Programmable Logic Arrays (PLAs) and Programmable Array Logics (PALs)



• PLA/PAL advantages

- Ease in circuit implementation (don't have to worry about placing and routing individual gates on a chip)
- The wiring is well-patterned and regular; thus easy to pre-estimate delay and area costs of the PLAs/PALs on a chip
- Disadvantage:
 - Not CMOS, so higher power consumption (esp. when o/p = 0)

A functional schematic; the implementation is more streamlined